Evening meeting:

The paper is to be presented at an evening meeting held at IStructE, 11 Upper Belgrave Street, London SW1X 8BH, on 4 November 2004, at 18.00h

Design and construction of Terminal 5 roof

Steve McKechnie
BSc, CEng, MIStructE
Arup

Paul Hulme
BEng
Watson Steel Structures Ltd

Gursharan Thind
BSc, MSc, DIC, CEng, MIStructE
Arup

Dervilla Mitchell
BE, CEng, MICE, MIEI
Arup

Keywords: Terminal 5, Heathrow Airport, London, Roofs, Design, Analysis, Steel, Fabricating, Construction Work

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Synopsis

This paper covers the evolution of the structural form and build sequence; the innovative node design; the analysis of tolerances and fit-up; the fabrication and castings; the temporary works and strand jacking; the behaviour during prestressing and load transfer and demonstrates how good design, construction and teamwork can bring additional value to our clients.

Introduction

Terminal 5 at Heathrow is a major new transport interchange sited at the western end of the airfield on the site of the former Perry Oaks sludge treatment works. The project has a long history: the first schemes were discussed in 1988; the project team was constituted in 1992; a public inquiry ran from 1995 to 1999; and the government gave the go ahead in November 2001.

Phase 1 of the development will increase capacity at Heathrow by 30M passengers per annum and will consist of two new terminal buildings, extensions to the Piccadilly line and Heathrow Express with associated stations, a new link to the M25, a large bus station and a multi-storey car park.

The scale and nature of this project presented major challenges. Rather than pursue a conventional approach BAA, the client, assembled a team working under a bespoke partnering contract: 'The T5 Agreement'. That document sets out how the parties would 'achieve success together and deliver world class performance'. It specifies:

- Collocated, integrated teams with all suppliers involved from the outset
- An emphasis on design for manufacture and assembly
- A focus on safety
- All financial and program risk ultimately remains with BAA
- Project insurance policies for PI, public liability and contractors all risks

'Partnering' and 'teamwork' are buzzwords in today's industry. They are often applied to situations where the people involved continue with the same old adversarial ways of working. We believe that the success of this project has been achieved through teamwork based on mutual respect, trust, and openness. This enabled Watsons to influence the design and gave Arup a say in the erection methodology. The architecture, the structural design and the erection strategy were all developed concurrently.

Structural form

Concepts

Airports are places of change, movement, of separations and new beginnings, dislocation, looking forward and looking back, pleasure, anticipation, waiting, frustration, and big boards with long lists of flight numbers.

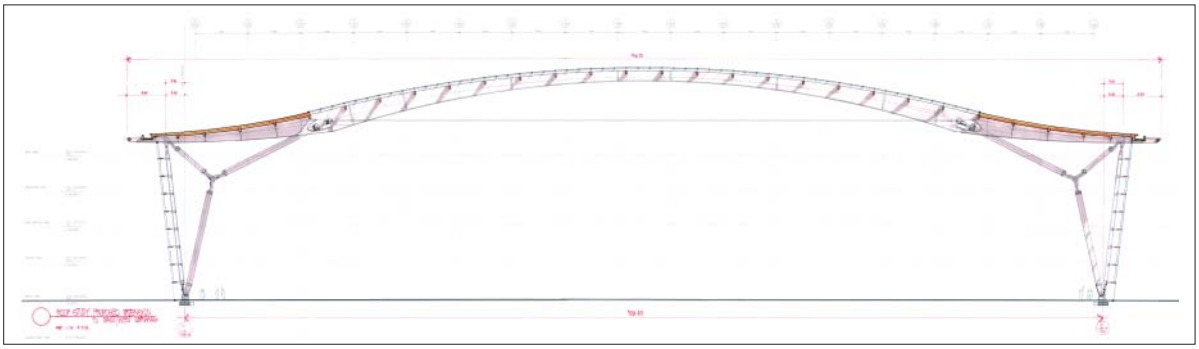
But despite the frenetic activity that will fill the terminal, BAA, and architects Richard Rogers Partnership, wanted to give the building a feeling of calm and unity. It should help people to find their place; both in the world, by having a strong identity, and in the terminal, by providing visual cues.

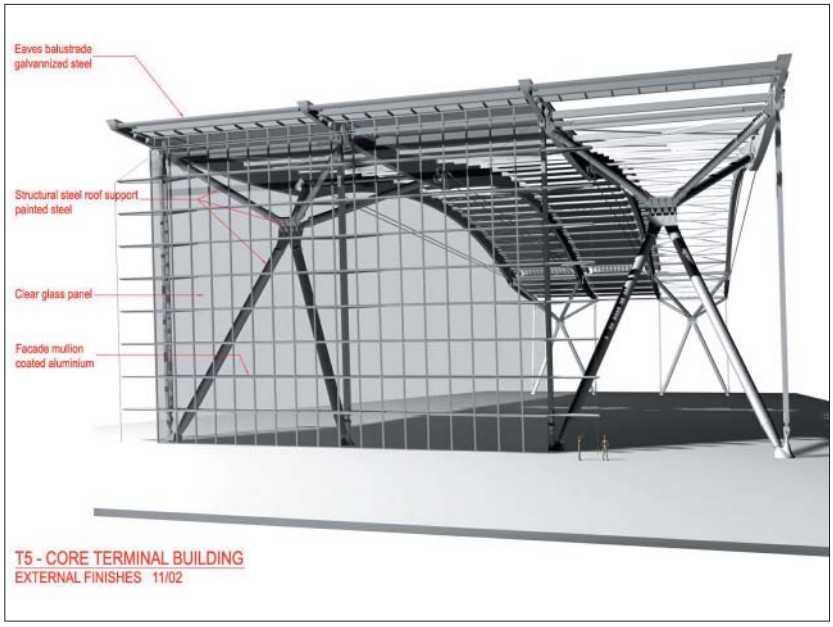
Crucially, BAA wanted to create a building that will have a place amongst the great transport interchanges of the world: The Gare du Nord, Grand Central Station, Paddington Station, and provide a memorable experience for travellers that will make them want to return time and again. This led them to choose a building with an exceptional architectural and structural vision: A single span roof that would vault over the superstructure of the entire building.

Schemes

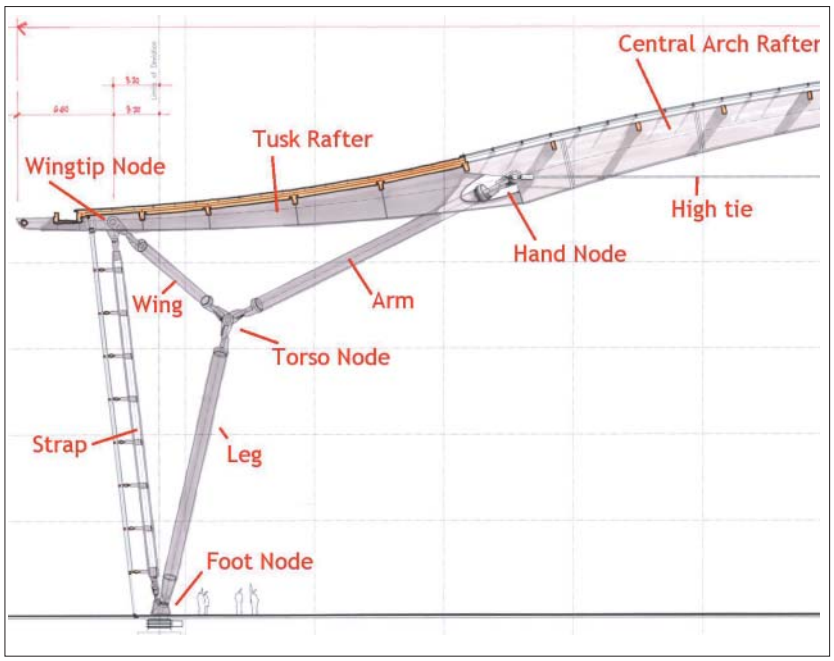
The design team considered many possible configurations of the structure. A truss or a pure beam would have had the advantage of not requiring any tying force, but beams turned out to be very heavy and trusses were visually too busy, detracting from that feeling of calm.

A very exciting early concept used a diagrid of tubular steelwork to form a folded plate shell 7.5m deep, spanning the whole 156m. Sadly the cost proved prohibitive and the site welding would have required 40 welders to be on site for 80 weeks.





(Left)
Roof cross section
(Below)
Design visualisation



Frame nomenclature

That made an arch or portal frame very much the front-runner. After studying the various possible fabrication methods the team decided on a hybrid structure with arched steel box girder rafters, 800mm wide and up to 3.8m deep. We adopted pairs of 115mm prestressed locked coil strand ties at high level to control spread of the arch and reduce midspan bending moments. The substructure has an 18m grid; the roof supports are therefore at 36m to allow for the departure gates and their circulation spaces. We inclined the compression elements of the support system (columns) and formed them into the 'cross bracing' to provide longitudinal stability.

The tension elements were kept vertical, adjacent to the façade, where they could also act as a direct support to the façade. They use the significant tensions that arise from their role in the roof system to help resist the moments that are generated by wind loads. This made it possible to give them a very slim profile that would only cause minimal visual intrusion into the views looking obliquely out through the façade.

We realised that once the centre section of roof was prestressed it would, in itself, form a spanning structure that would be capable of being assembled at low level and subsequently lifted into place. This became a key driver for the development of the erection method.

The secondary beams (purlins) span 18m so we made them 610mm deep steel box sections. The roofing is prefabricated into 'cassettes' that are lifted into place between the secondaries.

Analysis

Construction stages

The fact that the central arch section is loaded and prestressed before it is lifted into place means that the state of stress in the steelwork is very different from what you would get if you started with a complete frame and then applied the loads to it. We used GSA, the structural analysis program, to model the roof as a 3D skeletal frame. This allowed us to define 'construction stages', analyse the part completed frame and add the results to an analysis of the whole frame.

Each of the six 'lifts' of four rafters have to stand alone on the site for several weeks. The wind (and other) loads on this structure are very different to the loads that will be exerted on the completed building. So we analysed these part complete states in addition to a full half building model of the final state.

PΔ and buckling

The team were keen to make sure that we had fully accounted for PΔ effects and overall buckling. We did two things; firstly, we used second order analysis (PΔ) for all the ultimate limit state analyses and secondly, we made specific design allowances for the effects of possible initial imperfections in the erected structure. These allowances were established by generating a set of design forces that consisted of the directly applied forces added to the forces that would arise from the compressive forces acting on a structure that had initial imperfections built into it. The geometry and magnitude of these imperfections were derived from a series of buckling mode analyses in the GSA analysis program scaled to match the strut action moments that are implied by BS 5950, part 1, Annex C.

Wind load and deflections

The structural geometry of the roof means that it deflects significantly when loaded asymmetrically. In order to ensure that we had fully accounted for the asymmetrical effects of wind loading we carried out a time history dynamic analysis of the results from the wind tunnel test.

The wind tunnel laboratory (RWDI of Guelph, Ontario) instrumented one half of the T5A building model with 284 pressure taps and recorded pressure data in real time at a rate of 512 measurements per second. These results were passed to the Arup Advanced Technology Group (ATG) who converted them into a series of time varying loadings and ran a time history dynamic analysis using a version of the structural analysis model of the roof that they had converted to run in NASTRAN. They ran a data set that was equivalent to a 1-hour wind event and this was repeated for 8 different wind directions. This modelled, in real time, the way in which real turbulent wind flows would create varying pressures over the roof and the way that the roof would move in response to these pressures. Arup ATG also wrote post-processing software that abstracted critical deflection data, carried out structural utilisation checks and abstracted the critical force combinations for each member in the structure.

The deflections were significantly less in the time history analysis than had been suggested by our initial assumptions. As such, we were able to take 800te of steel out of the structure by making the rafter flanges and webs thinner. After the analysis had been re-run for this lighter frame, we recombined the element forces data that they produced with the data from the rest of the analysis so that we could use it for final member and connection design checks.

Presets

The team calculated preset values from the design models so that, after erection, the structure would end up in the position shown on the drawings. The central arched rafter was fabricated low and wide, ready to be pulled in at the ends during prestressing. The abutment CHSs were fabricated long, or short, depending on whether they were to be in compression or tension. The overall frame (abutments and central arch) was assembled high, ready for the downwards deflection during load transfer.

Lift 1 During temporary works slide

Tolerances and fit up

We made it a priority that the frame should be capable of being built to defined and yet achievable levels of dimensional accuracy without compromising the design or the programme. Our first step was to make a quantitative estimate of the effect that individual element tolerances would have on the dimensions of the overall frame. We then combined the results of all the possible deviations using the square root of the sum of the squares.

It became clear that the connection of the centre section to the abutments would need to be adjustable for both length and angle. We achieved this with TCBs (a proprietary HSFG bolt) in slotted holes in a traditional I section splice. In addition, the connection of the strap to the foot allowed the level of the eaves to be set independently of the position of the torso node.

Splice between centre section and abutment

(Left) Foot node

Connections

See 'Frame nomenclature' figure, page 26.

Torso node

Six large diameter CHS elements (914 x 40 and 711 x 50) node out at this point. The primary forces are axial, around 18000kN per CHS. Our first thought was to use a large steel casting; we also considered the use of a fabrication made up from welding plates or tubes. However, our welding engineer told us to be very cautious about the large and complex welds that would be required for the fabricated solution because any remedial works could be extensive and lead to programme delay. Furthermore, our metallurgist told us to be equally cautious about large castings since difficulties in prototyping could lead to delays late on during the design programme.

As a result of these warnings we developed a third option; nodes assembled from machined steel pieces that slot together. The load would be transferred from the leg into a thick plate. The two legs would share the same plate. The forces in the north-south direction thus would be resolved and we would do the same for the arms and the wings. The plates would be 250mm thick – it wasn't long before the team dubbed them 'megaplates'.

Where the three plates meet the force would be transferred via a pin to allow angular adjustment during erection. However, in order to fit a horizontal pin through a plate, the plate itself would have to be vertical and the pin section required to transfer 18000kN/leg would be huge. Therefore we used several interfaces on the pin to transfer the shear, reducing the pin diameter to 400mm. These shear planes would be provided by a series of vertical plates (the 'teeth') that would slot over the main megaplates.

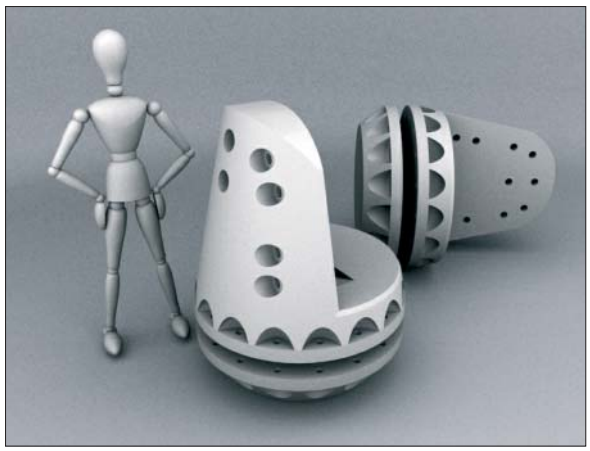
The ends of the tubes would be capped by a cast steel 'claw' that transfers the load to the megaplate via a cylindrically machined bearing surface. That would allow fit-up with angular adjustment in both directions and it would also bolt to the megaplate for robustness, and to carry self weight and strut action moments.

We carried out a study that compared the technical merits, risks, costs and programme implications of the three possible options. The welded fabrication was a clear loser from this process, while the castings and the megaplates had similar costs, but different risks and opportunities.

After much deliberation the team chose the megaplate solution because it offered:

- A wider range of possible suppliers.
- Detailing that was better suited to the node installation on site.
- A strong 'gusto' engineering aesthetic that would set the 'scale' and 'grain' of the building.

(Right) Torso node



(Above)
View of abutment showing hand and wingtip nodes
(Top right)
Rendering of claw casting from 'RhinoCERUS'
(Right)
Claw castings before removal of vent tubes

Other nodes
After this strategic decision had been taken on the torso node it became clear that we could use a similar approach for the design of the other two nodes. Pins could pass straight through the rafter section at the centroid of the hand and the wingtip, transferring load to reinforced sections of the webs. The north south components of the arm and wing forces are resolved at the node position by CHS tie members between the two hands or wingtips.

Megaplates
We specified S355 for the 250mm and 150mm plates, giving a p_y of 275 and 295N/mm^2 respectively. A fracture mechanics analysis showed that Charpy V notch toughnesses of 27J at –30°C and –20°C were required for the 250mm and 150mm plates respectively so we used S355 K2G3 and S355 J2G3. The standard method in BS 5950 cl2.4.4 would have required a test temperature of –90°C!

We specified a flatness tolerance of Class S (Special Flatness) and a thickness tolerance of Class C to BS EN 10029 to guarantee a minimum thickness of plate. This resulted in the plates having a positive tolerance of 3.2mm for the 150mm thick plates. We designed these values into the overall node dimensions so that the tooth plates, nominally 150 thick, are spaced at 154mm centres so that they do not clash with each other.

Castings
The basic raw material for any casting is scrap metal. It is crucial, therefore, to monitor closely the chemical content. We specified that the high strength castings should have a p_y of 413N/mm^2 and a Charpy V notch toughness of 27J at –30°C. An easy way to increase strength is to add more carbon but this reduces the ductility and causes problems with welding. Consequently, we specified a Carbon Equivalent Value of <0.49. William Cooks, the castings supplier, achieved these requirements by using steel with high nickel content since this adds strength but has only a limited effect on the CEV.

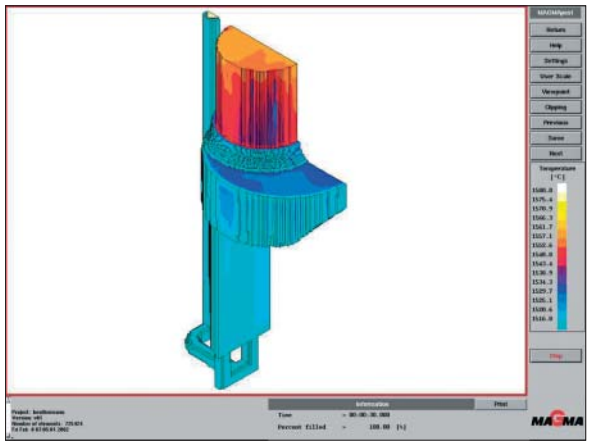
To verify the 'mix design', Cooks casted a sacrificial prototype of the largest high strength casting. It was destructively tested with test pieces cut from the critical structural sections. Good results were achieved for the strength but there were some low values for the toughness. We adjusted the heat treatment regime (quenching and tempering) and re-tested. This solved the problem. The resulting castings require a weld procedure with a pre-heat of 150°C applied via heating blankets and maintained for 12h before welding takes place.

RRP used 'RhinoCERUS', a 3D modelling package, to draw the initial geometry of the castings. Cross-sections were taken for use in the structural calculations and for inclusion on the 2D Autocad drawings. Critical structural setting out parameters, dimensions and tolerances were added to the '2D' drawings. We defined a permissible deviation of –0 / +10mm on cast surfaces. All connection surfaces were machined.

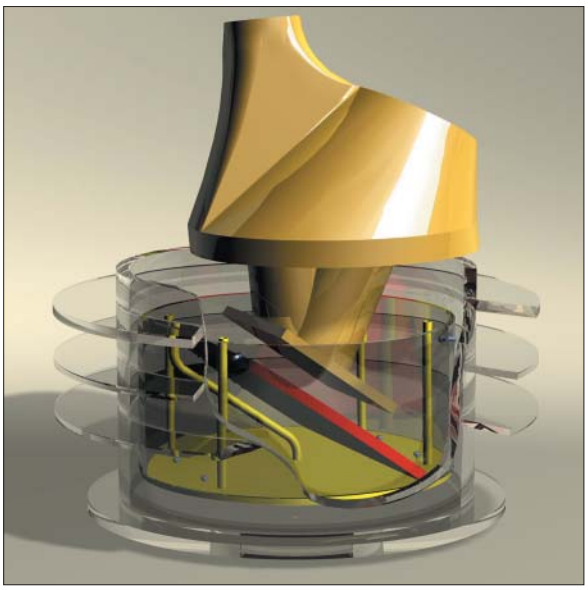
William Cooks proposed locations for the feeder head and vent tubes and where the split in the pattern would be best suited. As the visible surfaces of all the castings were to be 'as-cast' finishes it was crucial that no feeder heads or vent tubes would impinge on these locations. They also made allowances for shrinkage and tolerances before using the 3D 'Rhino' models to programme the CNC pattern cutting machines.

Before cutting any patterns, Cooks carried out a solidification simulation using 'Magmasoft'. This software gives an insight into the process of solidification that the casting will undergo as it cools. The aim is to eliminate defect spots and shrinkage cracks caused by differential cooling rates within the body of the casting as the melt solidifies.

Pins
Due to high in service stresses the pins at the centre of the nodes need a yield strength of 413N/mm^2 and the fracture mechanics requires a toughness of 27J at –30C. This is a very tough requirement for a 400mm thick section. It was decided to use forged steel and to specify a testing regime that used 100% MPI over the machined surface to ensure that there



An early Magmasoft run



Design visualisation of foot node and buckets

were no cracks, tears or planar indications over the surface that could give rise to crack propagation in service.

Foot node
As the name suggests this node connects the frame to the substructure. The node out point is 550mm below SSL, close to the centroid of the steel that frames the apron level (basement lid) slab. The roof steels will rotate measurably at this point during service, under wind and thermal loads, so we use a pot bearing set at an angle which is as near as possible to the line of combined axial force under all possible loading conditions.

In order to avoid any dependence on the positional accuracy of the substructure steel erection we put the pot bearing into an inner bucket (as it were) that fits into an 'outer bucket' with a 40mm grout layer between them.

Typical rafter splice
The rafter splices have saved a vast amount of site work compared with the more conventional welded alternative. The splices carry significant compressive loads; they are always in overall compression. As a result, the primary load transfer can be in bearing over the machined end faces of the flanges and webs. There are, of course, moments in both directions as well and so bolts are needed; fortunately, they fit inside the box section. Vertical shears are carried by friction between the bearing surfaces but horizontal shears and torsions are transferred through the two shear keys which also act as erection guides.

The positions of the splices were agreed very early on since they had to be coordinated with the temporary supports, which, themselves, had to be above substructure column positions.

Manufacture

Abutment first run study
BAA, recognising the complexity of the roof structure and the effect that its build duration would have on the overall T5 construction programme, backed the team's view that full scale trials should be carried out to prove the design, fit and assembly of the elements.

The most striking exercise that was undertaken was the building of one complete abutment structure at Dalton in North Yorkshire. This trial allowed the erection team to gain specific experience and highlighted some problems which were addressed prior to 'real' production; it also added significant impetus to the commencement of the production of the castings and nodes in earnest ahead of the project critical path.

Shop drawings
One major benefit of the AFRS was that it enabled detailed design issues to be resolved early and allowed us to commence the production of fabrication drawings with the necessary information.

Watson modelled the steel in 3D using XSteel. Presets and precambers were built into the XSteel model and so the job had to be split into sub models of the pre-assemblies. Special consideration was given to buildability; for example, special lifting brackets were designed for each major lift so that the piece would naturally swing into the correct attitude ready for connection and the splice plates on the hand were given a temporary hinge detail to avoid the need to lift the plates to high level.

Fabrication
The main fabrication centres were Watson Steel at Bolton and PPTH in Finland. William Cook's in Burton on Trent supplied the castings. Corus at Workington undertook the manufacture and assembly of nodes and the machining of the castings. Bridon supplied the high tie cables.

Manufacture was carried out using machined jigs to ensure dimensional accuracy of the complex parts.

The sourcing of materials was undertaken to minimise the need for butt-welding which would have been both costly and unsightly. This was achieved by arranging with the mills for the procurement of materials that were outside the normal mill-rolling catalogue.

(Below) Rafter splice showing female shear key

Erection and strand jacking

The project preference for prefabrication off-site, concerns regarding the availability of sufficient skilled labour and the limits on craneage due to the radar ceiling led the team to decide on an erection strategy which involved the construction of the central section of the roof at low level, strand jacking it into position and then connecting it to the abutment supports that had been built *in situ*.

The development of the erection scheme started with a detailed desktop study of the installation procedure. This helped to verify that the scope and interfaces between the work done by the various suppliers was fully co-ordinated.

As a result of this process, a 'site assembly manual' was put together. This comprehensive document contains a detailed sequence of the site operations that were agreed by all the team members before work started. It also defines, in detail, the roles and responsibilities of the site team. External audits were instigated to ensure that the procedures were being closely followed during the site operations.

There were many consequential benefits resulting from the production of the manual, for instance, it was realised that weather dependency could be taken out of the equation with a couple of relatively minor changes.

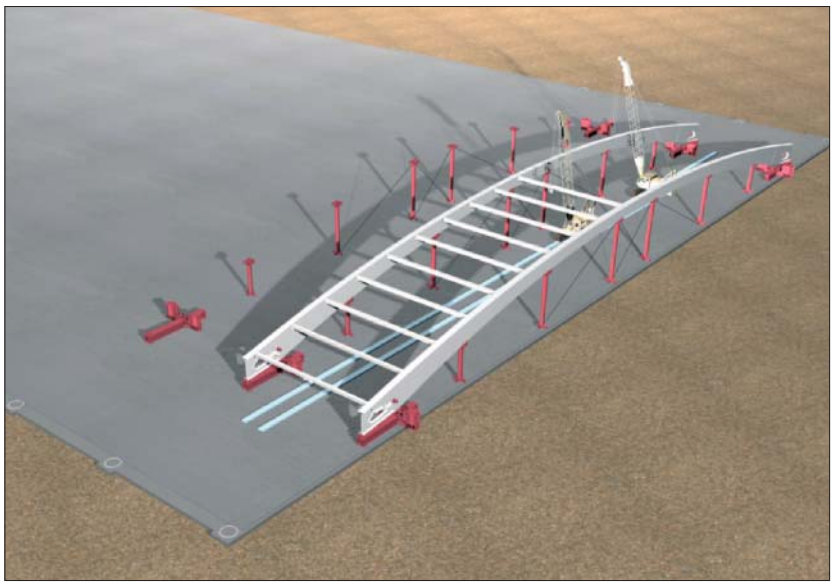
Temporary works
To carry out the scheme, a complex series of temporary works were required to support and manoeuvre the permanent works and also to enable the use of plant, both on the concrete slab of the supporting structure and around the perimeter of the building. Watson and The Rolton Group designed these. It would be possible to write a paper on these alone.

Here are just a couple of examples; the abutment temporary works, which could be described as a machine, rather than temporary works, provide 3 dimensional adjustments of the abutment permanent works and can be moved from phase to phase. A series of crane mats, that more resembled and acted as a bridge spanning between the steels of structure below, allowed the use of 70te crawler cranes on the top of the 200mm slab.

The build

Unfortunately it is not possibly to describe in detail each of the operations that have been developed and adopted in the build so we have summarised them in the following section.

The roof is constructed in six phases which are typically three bays wide (54m), one phase is a single bay of 18m.



Central arch section rafters being assembled on props

Centre arch section

The rafters are installed onto a series of temporary props, which are connected directly to the sub structure. The rafters, weighing up to 55t, and the secondaries are installed using the two 70t crawlers.

Starting at one end the rafters are connected sequentially via the shear key connection on their ends. The final section is installed using a retractable shear key.

The cladding system, a composite cassette system, is installed onto the roof module at low level and the remaining materials required to complete the cladding are loaded so that they get lifted, with the roof, up to high level.

Following the rafter build the high tie cables are installed. The cables are delivered in one length, rolled into a coil, placed on a turntable at one end of the rafters, unrolled over a series of rollers and linked to the dead end. The cable is then lifted sequentially onto supports and attached to the live end. The stressing equipment is installed and a nominal tension of around 10% of the design load induced.

A hand restraint truss is installed to provide torsional restraint to the rafters during cable tensioning.

The strand jacks are installed on top of 30m high temporary towers that sit on the main structure below the apron slab. There are two strand jacks, 180t capacity, at the end of each rafter.

Abutment structures

The pieces are craned into their final position supported by the temporary works and aligned using the jacking systems within it.

Four large crawler cranes are used. On each side of the building there is a 285t Liebherr LR1280 and a 450t Demag CC2500. The cranes run in a 30m wide crane track around the perimeter of the building. This area is specially prepared to withstand the large ground pressures exerted by the cranes, this includes the strengthening of the buried structures. Where crane access is restricted by excavations for the underground rail lines the cranes are rigged with additional jib length. Crane set-ups and moves are a significant determining factor for the programme.

Prestressing

The first part of the sequence involves lifting the ends of the rafters clear of their supports to eliminate friction from the stressing process. The dead load of the centre arch is transferred into the strand jacks incrementally. At each stage the verticality of the strand jack towers and structural integrity of both the temporary and permanent works is checked and signed off. The effect of the stressing induces compression into the rafter, which in turn deflects upwards along its length causing them to lift clear of the temporary supports.

The cables are tensioned incrementally in pairs. Initially, the cables are stressed in load increments until the rafters lift off the temporary supports. The stressing is then completed by increasing the load in order to limit the differential deflection of the rafters as they deflect upwards. This is monitored in real time using wireless instrumentation (a guy on a cherry picker with a radio) at each rafter location.

It should be noted that the cables are tensioned to a design load, not final length of span. Thus far the calculations have proved reliable and in every case the span achieved is well within the tolerances.

A final adjustment of the tensions in the cables is carried out after a further 12h to adjust for creep.

Lifting

The roof is lifted up into place using the strand jacks. The jacks are equipped with transducers that give an indication of lift load and stroke deflection but we discovered that to maintain levels of the rafter ends within the predefined limits an external measuring system is required. The final position that the roof is jacked to is preset higher than the final design position to allow for deflection during load transfer.

Connection and load transfer

In the final stage the remaining connections are completed and the load is released from the various jacking points, ending with the lowering away of the strand jacks. The final deflection is a combination of elastic deformation of the steel and the closing of gaps in the interfaces.

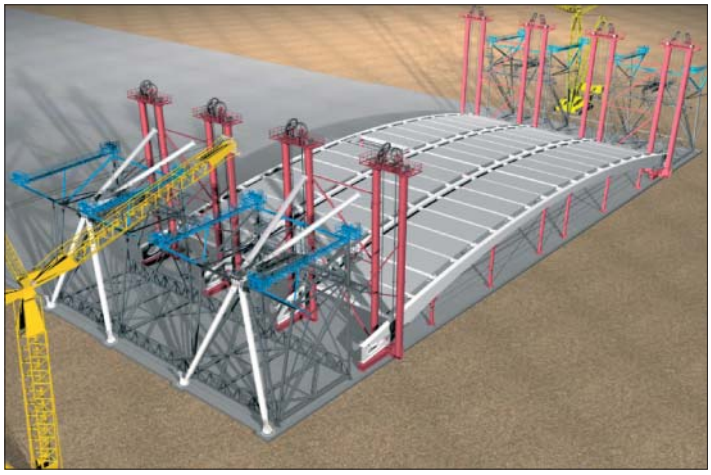
The load transfer is carried out in three stages.

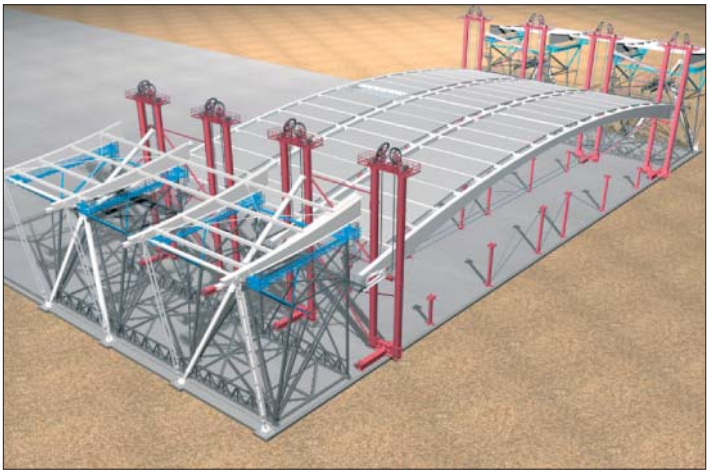
Firstly, the load is released from the abutment temporary works, leaving the structure supported on the strand jack towers and at the foot connection.

Secondly, the strand jacks are lowered until a shedding of load into the permanent foot support is observed. This indicates that the initial deflection due to gap closing has occurred.

Finally, the splice connection is completed and the remaining load released from the strand jacks.

(Below left) Abutment steel being placed on temporary works (Below right) The 'big lift'





(Left)
Temporary works in use for lift 2

Temporary works relocation

Following load transfer the temporary works can be relocated and the permanent structure completed.

The relocation has two stages: the strand jack towers are dismantled and moved piece small onto the next phase, whilst the abutment temporary works are launched along the slab. The frame has been designed to fit under the permanent works whilst moving to its next position. To eliminate loadings onto the concrete slab the structure is pulled using long stroke jacks over 'Hillman' rollers at main grid lines.

Once the structure has been moved to its new position the rollers are replaced with bearings and the process starts all over again.

Learning from lift 1 load transfer

Having the whole team involved in the construction means we can improve the process as we go along. For example, elastic analysis predicted that the frame would drop by 87mm during load transfer but on lift 1 the actual deflection was 182mm. We had expected the actual movements to be a little greater than predicted because of the impossibility of fully packing the arm to hand connection but this went far beyond that. We suspected that all the steel-to-steel connections had been fitted together with a small gap remaining at each interface and so we ran a structural model in which each member was shortened by 1mm for each steel-to-steel interface. This produced a pattern of deflection that was very similar to the additional deflections. We used this model to estimate the forces generated by the additional deflections. We checked the reserve of strength in the structure and, happily, it was great enough to carry the additional forces.

Improved load transfer procedure

The first lift dropped by an extra 95mm so we preset the second lift by an extra 90mm upwards. In order to prevent the build up of additional moments in the rafter the TCB bolts in the top flange and web of the tusk rafter splice remained un-torqued for the first 90mm of the load transfer. This allowed the roof weight to close the gaps at the bearing surfaces. The splice bolts were then torqued and the rest of the load transferred. This worked well. The load readout on the strand jacks indicated that for the first 70mm of the load transfer very little load was transferred. The load increased to 25% transfer by the 90mm point. By 100% load the roof had dropped by around 80mm further than the elastic prediction.

Conclusion

This paper is littered with examples of how the integrated team has woven together the requirements of architecture, fabrication, erection and structural design to produce a momentous piece of engineering. BAA are pleased with the financial safety and programme performance of the roof team as well as with the 'product' itself. This is a result of good design and good construction by a team that has worked together with but one single aim; the creation of a great building.

IStructE

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